

A Full Affirmative Answer to the Fractional-Range Question for Cyclic Projections

Literature-implied answer packet for arXiv:1510.04560
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Status: literature-implied answer (full). For every finite family of closed subspaces and every $0 < \alpha < 1/2$, the algebraic sum of their orthogonal complements is contained in $\text{Ran}(I - T)^\alpha$. Thus the question in Badea–Seifert has a full affirmative answer, with a uniform interval of exponents and without the aligned-case hypothesis. The proof combines Reich–Zalas’s later operator estimate with the binomial functional calculus. This implication appears not to have been stated in the supporting paper and is identified here.

1 Source question

Let $M_1, \dots, M_N$ be closed subspaces of a real or complex Hilbert space X , let $P_i = P_{M_i}$, and set

$$T = P_N \cdots P_1.$$

Badea and Seifert [1, Remark 4.4(b), PDF p. 14] ask, in the aligned case, whether

$$M_1^\perp + \cdots + M_N^\perp \subseteq \text{Ran}(I - T)^\alpha \tag{1}$$

for some $\alpha \in (0, 1)$, or at least whether the left-hand side is contained in $\bigcup_{\alpha > 0} \text{Ran}(I - T)^\alpha$. They note that either inclusion would imply a conjecture of Deutsch and Hundal concerning the location of vectors witnessing arbitrarily slow cyclic-projection convergence.

for all $x \in X$. If the subspaces $M_1, \dots, M_N$ are not aligned, then the convergence is exponentially fast and in fact

$$(4.3) \quad \|x_n - P_M x\| \leq \left(1 - \frac{3(N-1)}{N^3}(1-c)\right)^{n/2} \|x - P_M x\|, \quad n \geq 0,$$

for all $x \in X$, where $c = c(M_1, \dots, M_N)$. On the other hand, if the subspaces $M_1, \dots, M_N$ are aligned, then the convergence is arbitrarily slow but for each $\alpha > 0$ there exists a dense subspace X_α of X such that

$$\|x_n - P_M x\| = o(n^{-\alpha}), \quad n \rightarrow \infty,$$

for all $x \in X_\alpha$. Furthermore, there exists a dense subspace X_∞ of X such that for all $x \in X_\infty$ the convergence in (4.2) is super-polynomially fast.

Proof. Let $T \in \mathcal{B}(X)$ be given by $T = P_N \cdots P_1$, so that $x_n = T^n x$ for all $n \geq 0$. By Proposition 4.1 $W(T)$ lies in a Stolz domain, and the proof of that result shows that $\text{Fix}(T) = M$. Since $\text{Ran}(I - T)$ is closed if and only if the subspaces are not aligned by [4] Theorem 4.1], the result follows immediately from Corollaries 2.6 and 3.4. $\square$

Remark 4.4. (a) The estimate in (4.3) can be sharpened slightly to an estimate involving the inner ℓ^2 -inclination rather than the Friedrichs number by replacing the application of Corollary 3.4 in the above proof with an application of Theorem 3.3; see also Remark 4.2(a).

(b) As was noted in the introduction, exponentially fast convergence in (4.2) is equivalent to the subspace $M_1^\perp + \cdots + M_N^\perp$ being closed in X ; see [22] Theorem 6.4] and also [4] Theorem 4.1] for a number of further equivalent conditions. Thus both $M_1^\perp + \cdots + M_N^\perp$ and $\text{Ran}(I - T)$ are closed if and only if $c(M_1, \dots, M_N) < 1$. In fact, the identity

$$(4.4) \quad I - T = \sum_{k=1}^N (I - P_k) \prod_{j=1}^{k-1} P_j$$

implies that $\text{Ran}(I - T) \subset M_1^\perp + \cdots + M_N^\perp$, and in particular both spaces are dense in $M^\perp$. However, the inclusion is in general strict when $c(M_1, \dots, M_N) = 1$. Indeed, suppose that $N = 2$ and that $M_1 \cap M_2 = M_1^\perp \cap M_2^\perp = \{0\}$. Then the assumption that $(I - P_2)(M_1^\perp) \subset \text{Ran}(I - T)$ would already imply, by (4.4), that for each $x \in X$ there exists $y \in X$ such that

$$(I - P_2)((I - P_1)x - P_1 y) = (I - P_1)y.$$

But then both sides must lie in $M_1^\perp \cap M_2^\perp = \{0\}$, and therefore $(I - P_1)x - P_1 y \in M_2$. It follows that $M_1^\perp \subset M_1 + M_2$, and hence $M_1 + M_2 = X$. In particular, $M_1 + M_2$ is closed, and it follows from [19] that $c(M_1, M_2) < 1$. We do not know whether $c(M_1, \dots, M_N) = 1$ implies that $M_1^\perp + \cdots + M_N^\perp \subset \text{Ran}(I - T)^\alpha$ for some $\alpha \in (0, 1)$ or more generally that $M_1^\perp + \cdots + M_N^\perp \subset \bigcup_{\alpha > 0} \text{Ran}(I - T)^\alpha$; see also Remark 2.5(a). A positive answer to either of the above would imply a positive answer to the conjecture made in [22] Remark 6.5(2)].

Corollary 4.5. *Given $N \geq 2$ closed subspaces $M_1, \dots, M_N$ of a Hilbert space X , let P_k denote the orthogonal projection onto M_k , $1 \leq k \leq N$.*

Figure 1: Badea–Seifert, arXiv:1510.04560, PDF page 14. The fractional-range question appears near the bottom of Remark 4.4(b).

2 Later estimate and the answer

Write $Q_i = P_{M_i^\perp} = I - P_i$. Reich and Zalas prove the following operator-norm estimate [2, Lemma 4.1(iii), PDF p. 10]:

$$\max_{1 \leq i \leq N} \|T^n Q_i\| = O(n^{-1/2}). \quad (2)$$

Their paper is written for real Hilbert spaces. For a complex Hilbert space, apply the estimate to its underlying real Hilbert space; the projections, products, and operator norms are unchanged.

Theorem 1. *For arbitrary closed subspaces $M_1, \dots, M_N$ of a real or complex Hilbert space and $T = P_N \cdots P_1$, one has*

$$M_1^\perp + \cdots + M_N^\perp \subseteq \text{Ran}(I - T)^\alpha \quad \text{for every } 0 < \alpha < \frac{1}{2}.$$

More precisely, for each i there is a bounded operator B_i such that

$$(I - T)^\alpha B_i = Q_i. \quad (3)$$

Proof. Fix $0 < \alpha < 1/2$. Let

$$c_n = \frac{\Gamma(n + \alpha)}{\Gamma(\alpha)\Gamma(n + 1)}, \quad n \geq 0,$$

so that $(1 - z)^{-\alpha} = \sum_{n \geq 0} c_n z^n$ for $|z| < 1$ and $c_n = O((n + 1)^{\alpha-1})$. For each i , define

$$B_i = \sum_{n=0}^{\infty} c_n T^n Q_i. \quad (4)$$

By (2), this series converges absolutely in operator norm, since

$$\sum_{n=0}^{\infty} c_n \|T^n Q_i\| \lesssim 1 + \sum_{n=1}^{\infty} n^{\alpha-3/2} < \infty.$$

On the other hand, the standard binomial functional calculus for the contraction T gives the absolutely convergent series

$$(I - T)^\alpha = \sum_{m=0}^{\infty} d_m T^m, \quad d_m = (-1)^m \binom{\alpha}{m}. \quad (5)$$

Indeed, $|d_m| = O(m^{-\alpha-1})$. The product of (4) and (5) is an absolutely convergent double series: because T is a contraction,

$$\sum_{m,n \geq 0} |d_m| c_n \|T^{m+n} Q_i\| \leq \left(\sum_{m \geq 0} |d_m| \right) \left(\sum_{n \geq 0} c_n \|T^n Q_i\| \right) < \infty.$$

We may therefore take its Cauchy product. The scalar generating functions satisfy $(1 - z)^\alpha (1 - z)^{-\alpha} = 1$, and hence

$$(I - T)^\alpha B_i = \sum_{k=0}^{\infty} \left(\sum_{m+n=k} d_m c_n \right) T^k Q_i = Q_i.$$

This proves (3). Finally, if $x = x_1 + \cdots + x_N$ with $x_i \in M_i^\perp$, then

$$x = \sum_{i=1}^N Q_i x_i = (I - T)^\alpha \left(\sum_{i=1}^N B_i x_i \right),$$

which proves the asserted range inclusion. $\square$

Proof. Let $z \in C$. Knowing that $P_{C_m} \dots P_{C_1}$ is $(1/m)$ -strongly quasi-nonexpansive, we have

$$\|y_{k+1} - y_k\|^2 \leq m(\|y_k - z\|^2 - \|y_{k+1} - z\|^2). \quad (3.14)$$

see, for example, [14, Corollary 4.5.3]. Consequently,

$$\sum_{k=1}^{\infty} \|y_k - y_{k-1}\|^2 \leq m\|y_0 - z\|^2 < \infty. \quad (3.15)$$

In particular, $\sum_{n=k}^{\infty} \|y_n - y_{n-1}\|^2 \rightarrow 0$ as $k \rightarrow \infty$. By (3.7), we have

$$\frac{k}{2} \|y_k - y_{k-1}\|^2 \leq \left\lceil \frac{k}{2} \right\rceil \|y_k - y_{k-1}\|^2 \leq \sum_{n=\lceil k/2 \rceil + 1}^k \|y_n - y_{n-1}\|^2 \rightarrow 0 \quad (3.16)$$

as $k \rightarrow \infty$. This proves (3.12). The rate of (3.13) follows immediately from Lemma 3.1. $\blacksquare$

4 Closed and Linear Subspaces

In this section we oftentimes use the product space notation introduced in Section 2. The following result is a direct consequence of Theorem 1.4, Lemma 2.3 and Lemma 3.1.

Lemma 4.1. *For the operators T defined in (1.2) and $\mathbf{T}$ defined in (2.8), we have:*

$$\begin{aligned} (i) \quad & \|T^k - T^{k-1}\| = \mathcal{O}(k^{-1}); & (vi) \quad & \|(\mathbf{T}^k - \mathbf{T}^{k-1})P_{\mathbf{D}}\| = \mathcal{O}(k^{-1}); \\ (ii) \quad & \max_{i=1,\dots,m} \|P_{M_i^\perp} T^k\| = \mathcal{O}(k^{-1/2}); & (vii) \quad & \|P_{\mathbf{C}^\perp} \mathbf{T}^k P_{\mathbf{D}}\| = \mathcal{O}(k^{-1/2}); \\ (iii) \quad & \max_{i=1,\dots,m} \|T^k P_{M_i^\perp}\| = \mathcal{O}(k^{-1/2}); & (viii) \quad & \|\mathbf{T}^k P_{\mathbf{D}} P_{\mathbf{C}^\perp}\| = \mathcal{O}(k^{-1/2}); \\ (iv) \quad & \max_{i=1,\dots,m} \|(T^k - T^{k-1})P_{M_i^\perp}\| = \mathcal{O}(k^{-3/2}); & (ix) \quad & \|(\mathbf{T}^k - \mathbf{T}^{k-1})P_{\mathbf{D}} P_{\mathbf{C}^\perp}\| = \mathcal{O}(k^{-3/2}); \\ (v) \quad & \max_{i,j=1,\dots,m} \|P_{M_i^\perp} T^k P_{M_j^\perp}\| = \mathcal{O}(k^{-1}); & (x) \quad & \|P_{\mathbf{C}^\perp} \mathbf{T}^k P_{\mathbf{D}} P_{\mathbf{C}^\perp}\| = \mathcal{O}(k^{-1}). \end{aligned}$$

Proof. Statement (i) follows from Theorem 1.4 and the uniform boundedness principle [13, Theorem 2.2]. In view of Lemma 2.3, it suffices to show statements (vii)–(x).

(vii). We show that

$$\|P_{\mathbf{C}^\perp} \mathbf{T}^k P_{\mathbf{D}}\| \leq \sqrt{\frac{m}{2}} \|(\mathbf{T}^k - \mathbf{T}^{k-1})P_{\mathbf{D}}\|. \quad (4.1)$$

Let $\mathbf{x} = (x_1, \dots, x_m) \in \mathbf{H}$. Moreover, let $\{y_k\}_{k=0}^\infty$ be defined by the method of cyclic projections (1.1) with $y_0 := \frac{1}{m} \sum_{i=1}^m x_i$. Then, using the coordinate-wise projection formulas

Figure 2: Reich–Zalas, arXiv:2205.13843, PDF page 10. Lemma 4.1(iii) is the operator estimate used in the factorization.

3 Consequences, scope, and provenance

The theorem answers the stronger first alternative in the source question. It also gives more than was requested: the inclusion holds for every $\alpha \in (0, 1/2)$, and no alignment assumption is needed. In particular, every vector in $M \oplus (M_1^\perp + \cdots + M_N^\perp)$, where $M = \bigcap_i M_i$, has at least the polynomial convergence supplied by the fractional-power theory. Reich and Zalas directly prove the sharper orbit bound $O(n^{-1/2})$ on this subspace in their Theorem 4.3.

The strict inequality $\alpha < 1/2$ is exactly what makes the norm series (4) summable from the known $n^{-1/2}$ estimate. The argument does not settle the endpoint $\alpha = 1/2$, but the source question only asks for one exponent in $(0, 1)$.

A bounded search using the arXiv identifiers, titles, exact range notation, and the core cyclic-projection keywords located no paper explicitly stating this factorization or the resulting answer to Remark 4.4(b). The supporting paper proves (2) and the orbit-rate consequences, not the fractional-range inclusion. Accordingly, this packet is classified as a *literature-implied* full answer and makes no claim that Reich and Zalas presented it as an answer to Badea–Seifert.

References

- [1] C. Badea and D. Seifert, *Ritt operators and convergence in the method of alternating projections*, J. Approx. Theory 205 (2016), 133–148; arXiv:1510.04560.
- [2] S. Reich and R. Zalas, *Polynomial estimates for the method of cyclic projections in Hilbert spaces*, Numer. Algorithms 94 (2023), 1217–1242; arXiv:2205.13843, doi:10.1007/s11075-023-01533-w.